



CHAPTER 3 - MOBILE DESIGN PRINCIPLES

Topic Outline



1. Fitt's Law in Mobile App Design
2. Designing Interaction
3. Designing for Context

FITT'S LAW IN MOBILE APPLICATION DESIGN

Introduction

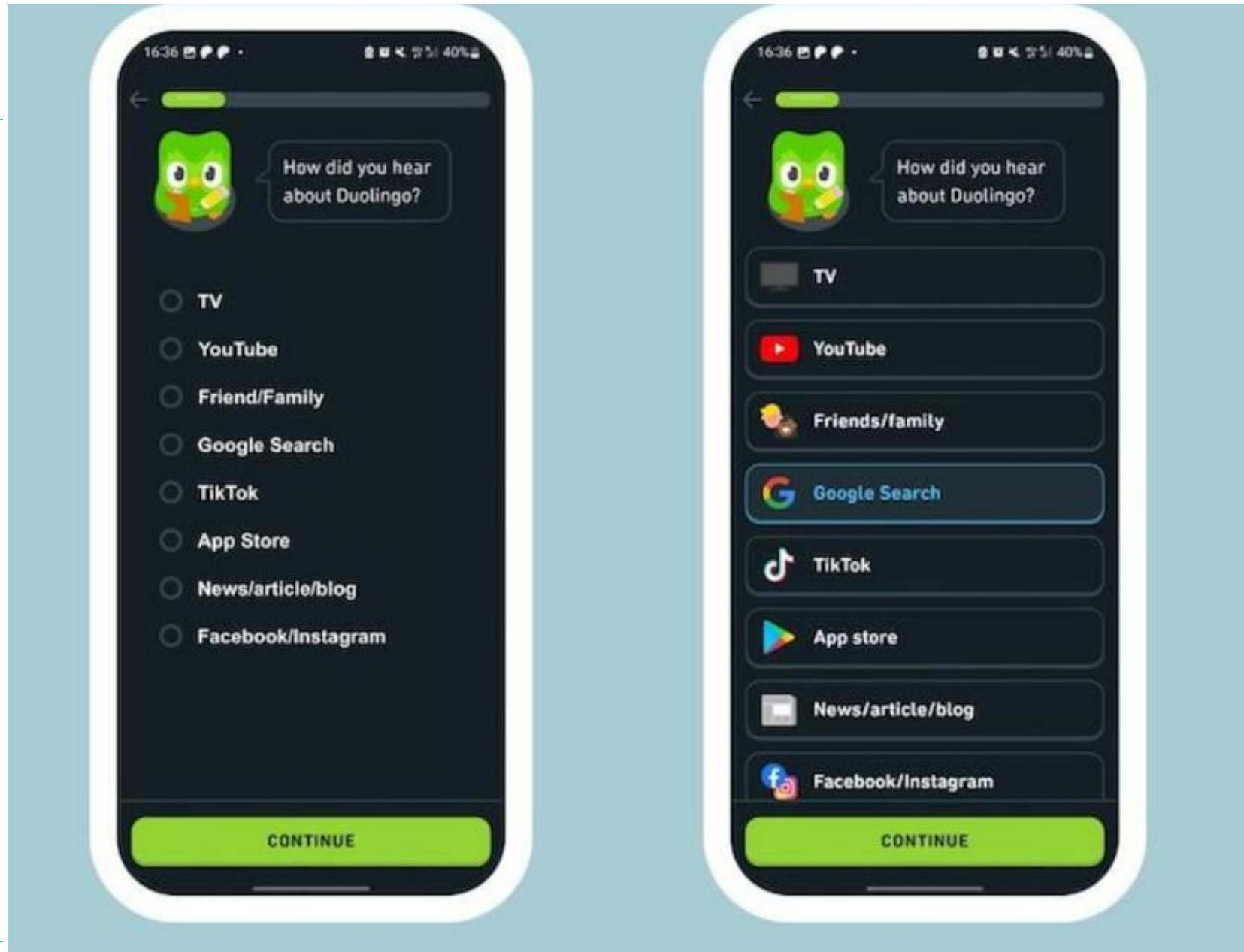


- ▶ Fitt's Law is a principle in **psychology within Human-Computer Interaction** which states the time it takes to move to a target depends on the **distance to the target** and the **target size**
- ▶ Thus
 - ▶ The **closer the target** and the **larger the target**, would make it **easier** for the user to interact.
 - ▶ **Larger targets** reduce movement time
 - ▶ **Shorter distances** reduce movement time
 - ▶ Both **size** and **placement** affect perceived effort



Fitt's Law in Mobile Design

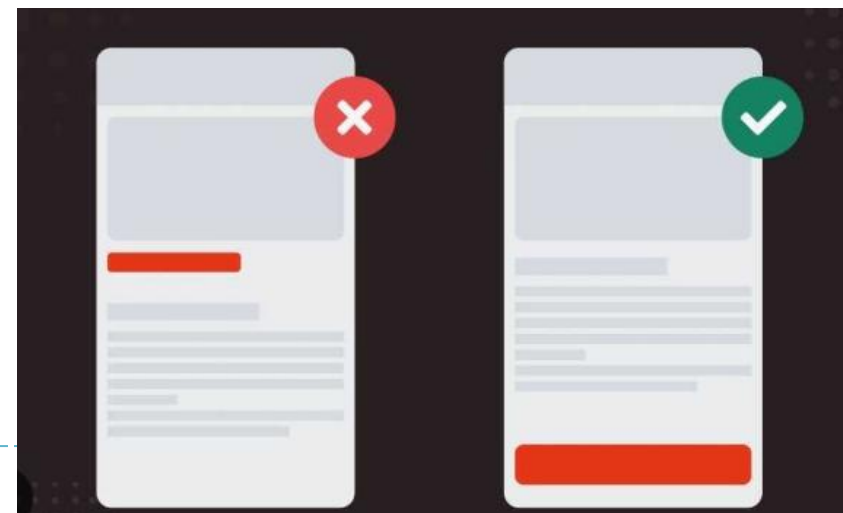
- ▶ Fitt's Law implementation in Mobile UI is crucial.
- ▶ This is due to the **screen size limitation** and the “**Easily Distracted**” nature of mobile user.
- ▶ Touch targets should be large enough and placed where fingers naturally rest.
- ▶ Buttons in corners are easier to target due to natural grip and thumb reach.
- ▶ Buttons in center should span screen width to reduce movement time.





Fitt's Law in Mobile Design Rule of Thumb

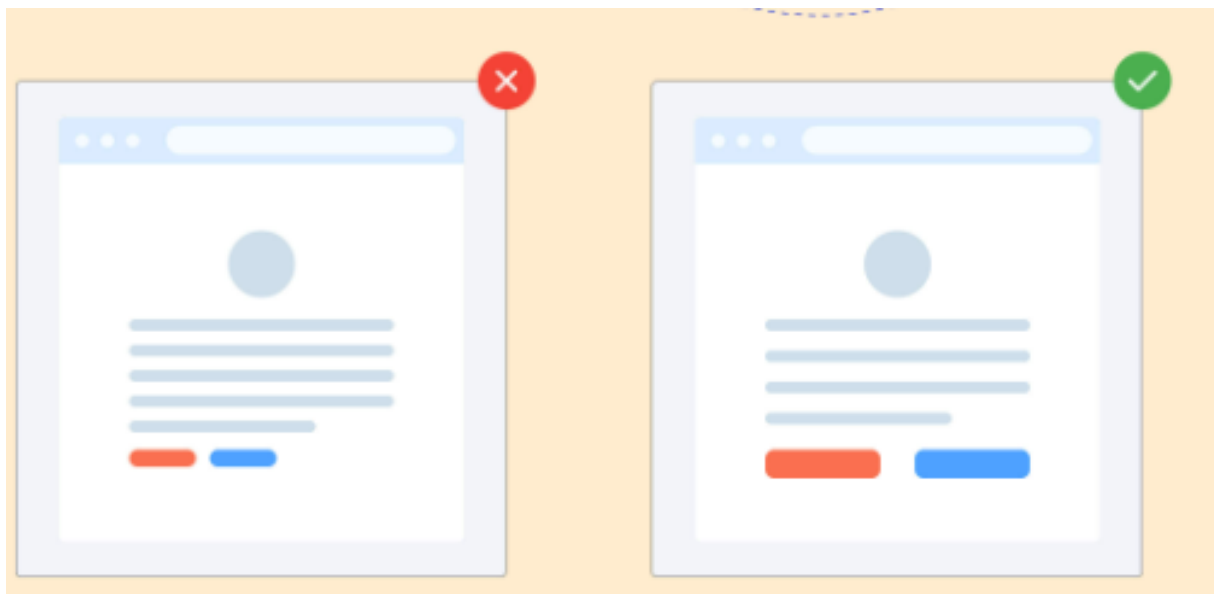
- ▶ Place **overflow button or menu buttons at the edges of the screen**
- ▶ must be as wide as the mobile device screen size (with an appropriate margin)
- ▶ Always prioritize placing **action buttons at the bottom of the screen.**
- ▶ **NEVER** place buttons or UI elements at the center of the screen with its width less than half of the screen width.





Fitt's Law in Mobile Design Rule of Thumb

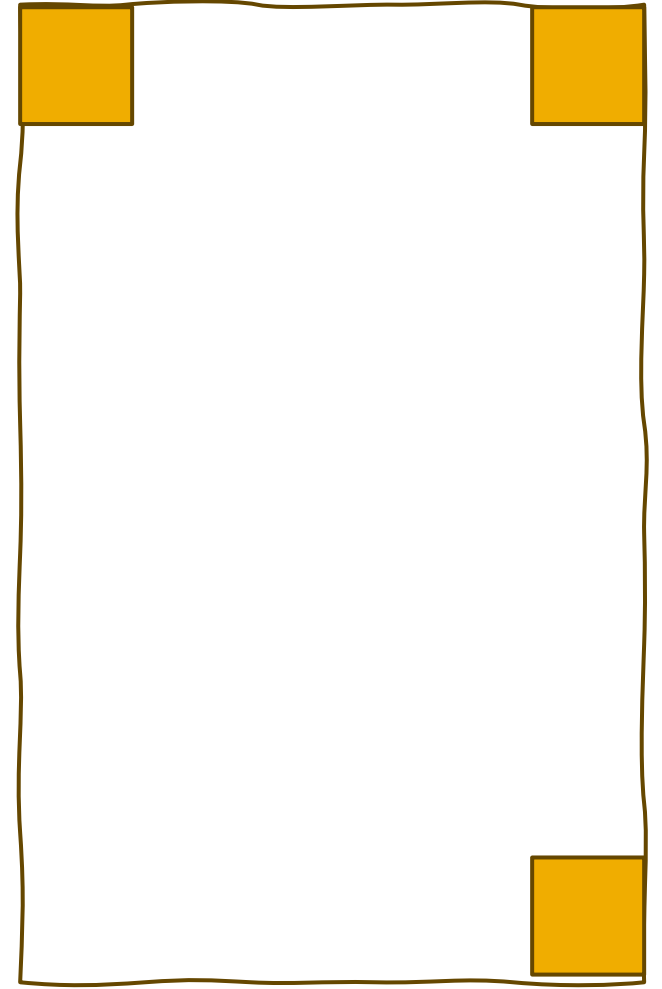
- ▶ If you must place buttons side-by-side, ensure the **total width of the buttons is equal to the screen width**
- ▶ Only put two text button **side by side**
- ▶ If you must put more than two buttons, then **use graphical buttons**



Fitt's Law Hot Corners in Mobile Applications



- ▶ Buttons in corners are easier to target due to natural grip and thumb reach.
- ▶ **In most mobile application, can you tell which buttons that are usually present at these corners?**





In Summary

- ▶ Fitt's Law explained that:
 - ▶ **Larger targets** reduce movement time
 - ▶ **Shorter distances** reduce movement time
 - ▶ Both size and placement affect perceived effort
- ▶ Call of actions buttons should be placed at the bottom
- ▶ UI elements that are not at the corners must be as wide as the screen width.
- ▶ Buttons which leads to additional options (Menus/Navigational Drawers) should be placed at the corners.



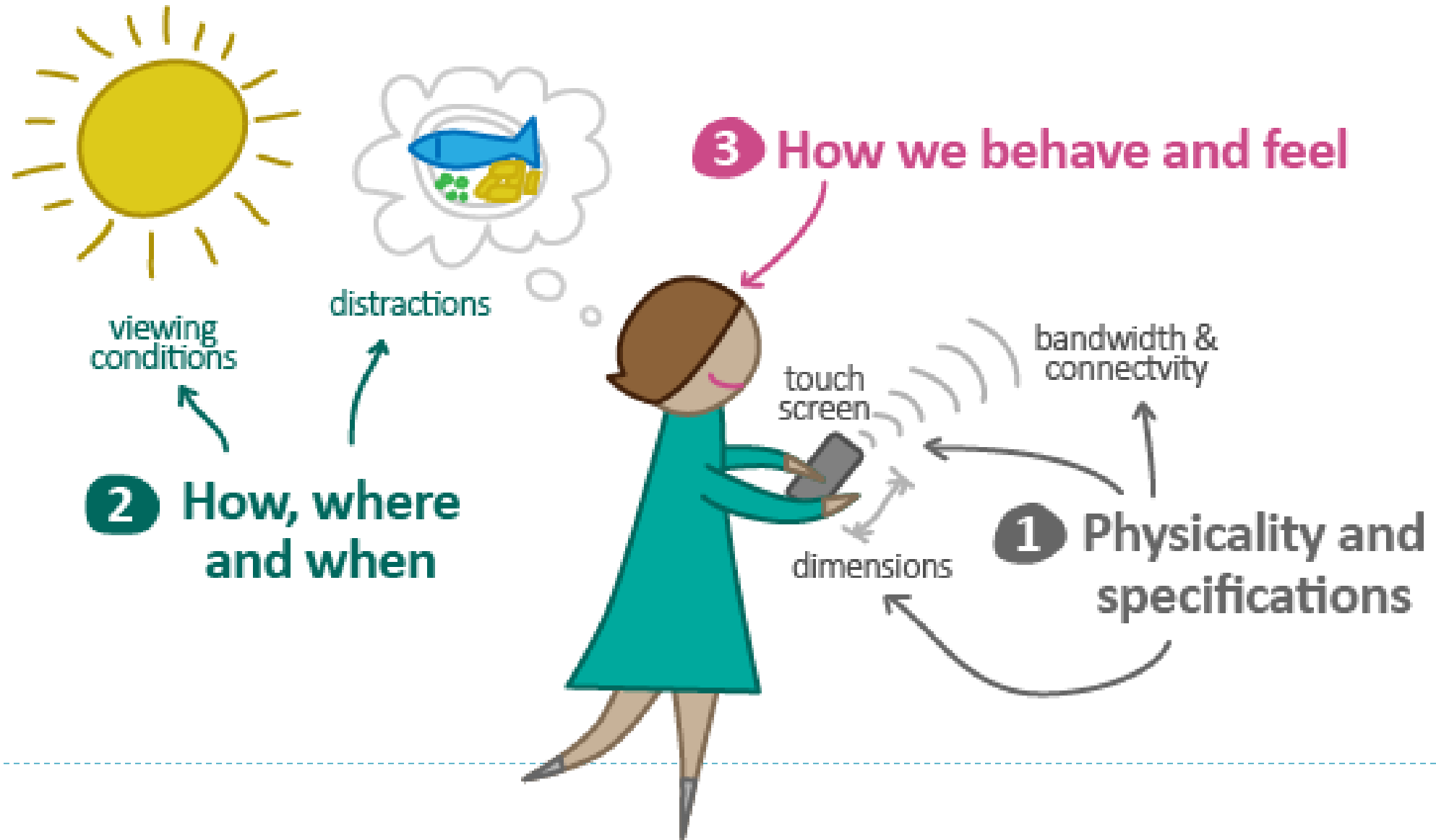
DESIGNING INTERACTION IN MOBILE



Do you feel any emotional connection to your phone?



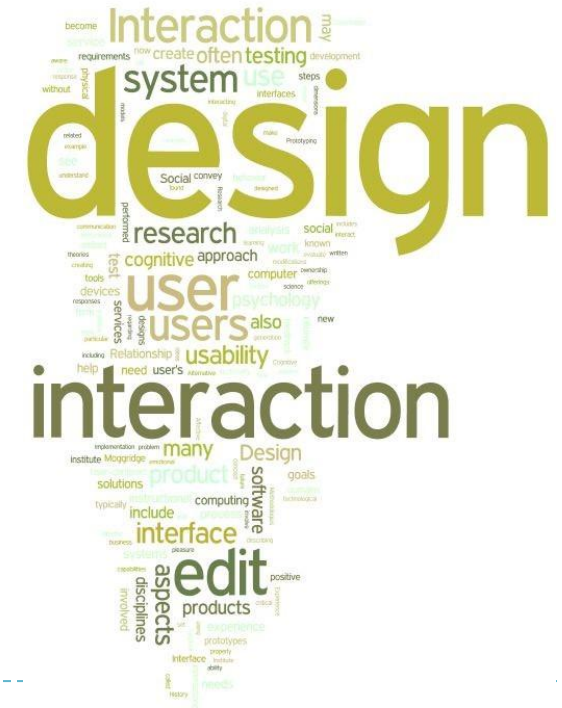
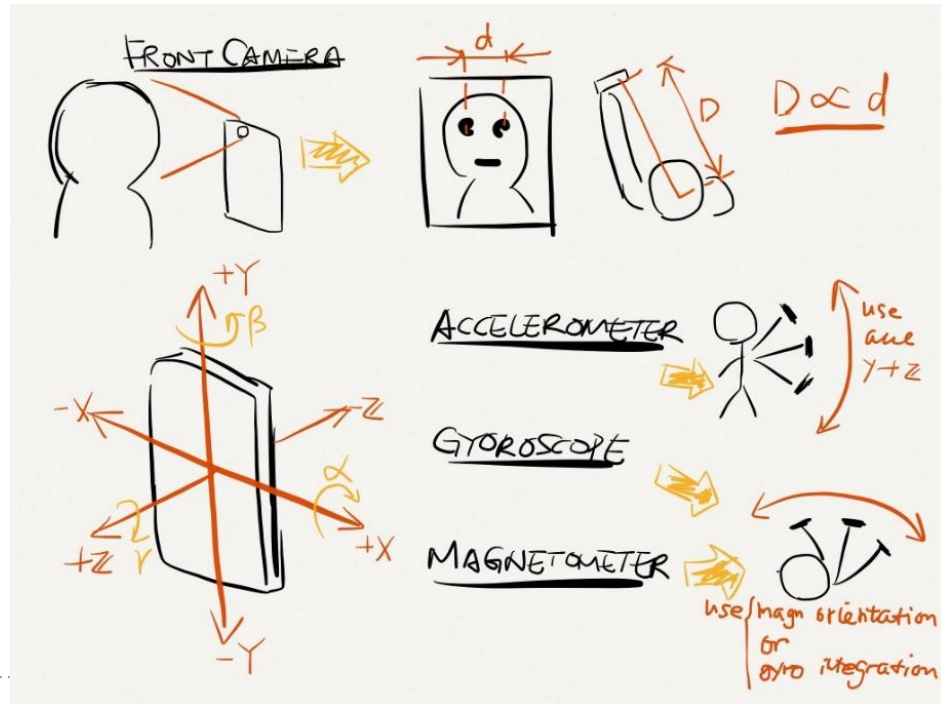
How is mobile different?



Interaction Design



- ▶ The design of how the user can **participate** with the information present, either in a direct or indirect way, meaning how the user will **interact** with the website or application to create a more **meaningful experience** and accomplish his/her goals.



Designing Interaction in Mobile



- ▶ When designing interactions, we should consider:
 - ▶ Most modern mobile devices employ touch screens which provides **opportunities** and **constraints**
 - ▶ The design should not only emphasis on content, but **how users would interact with the content**





Designing Interaction in Mobile Web

When Designing for ergonomics, one should consider:

1. Device dimensions
2. User holding patterns
3. Hit areas
4. Information Architecture

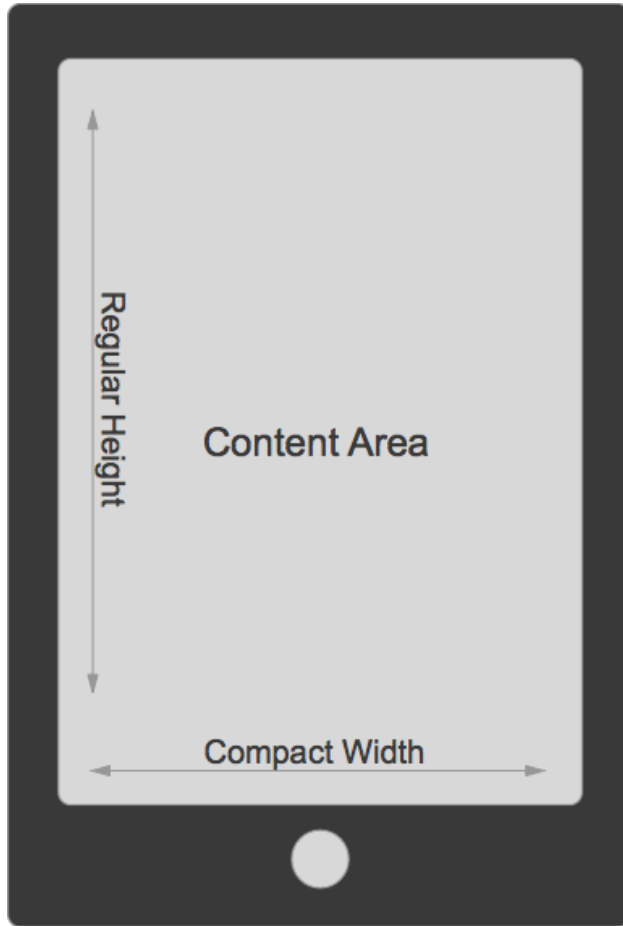
Device Dimensions



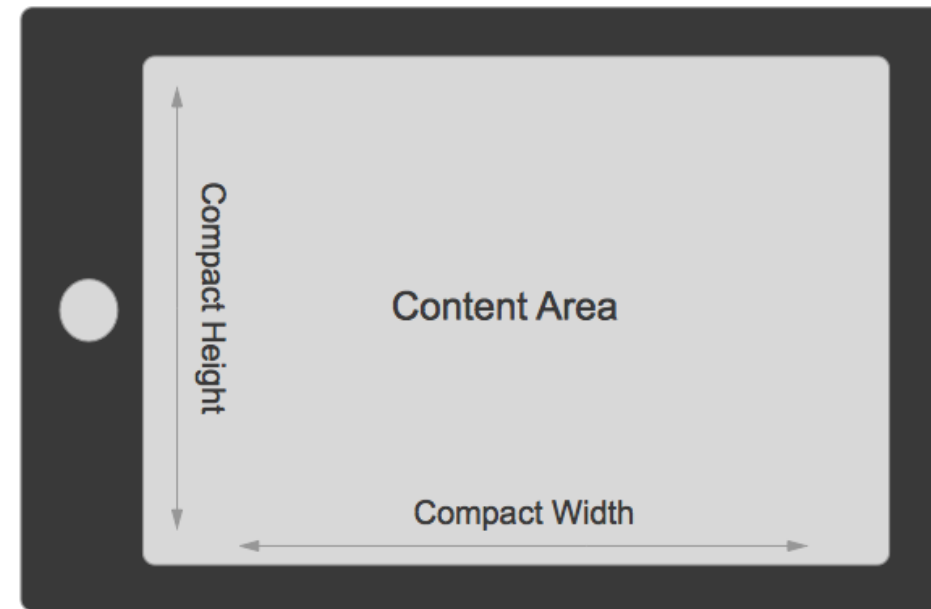
- ▶ A good mobile developers would take account into designing for different device dimensions
- ▶ There are several popular **aspect ratios** for mobile devices and **two main orientations**.



Popular Screen Orientation



iPhone Portrait View



iPhone Landscape View

User Holding Pattern



- ▶ According to Hooper (2013), users are found to held their phones in **three** basic ways:
 - ▶ One Handed
 - ▶ Cradled
 - ▶ Two handed
- ▶ However, the way in which users hold their phone is **not a static**. Users **change** the way they're holding their phone very often—sometimes every few seconds.
- ▶ This usually **relate to the task** they are performing while on mobile device.

Thumb Zone



- ▶ The Thumb Zone is a heat map of sorts.
- ▶ It's a best guess for how easy it is for our thumbs to tap areas on a phone's screen.
- ▶ Most common use case:
 - ▶ one-handed use
 - ▶ right thumb on the screen
 - ▶ thumb anchored in the lower-right-hand corner

Thumb-Friendly Heat Map Analysis



NATURAL

Easy to reach
area of the
screen

STRETCH

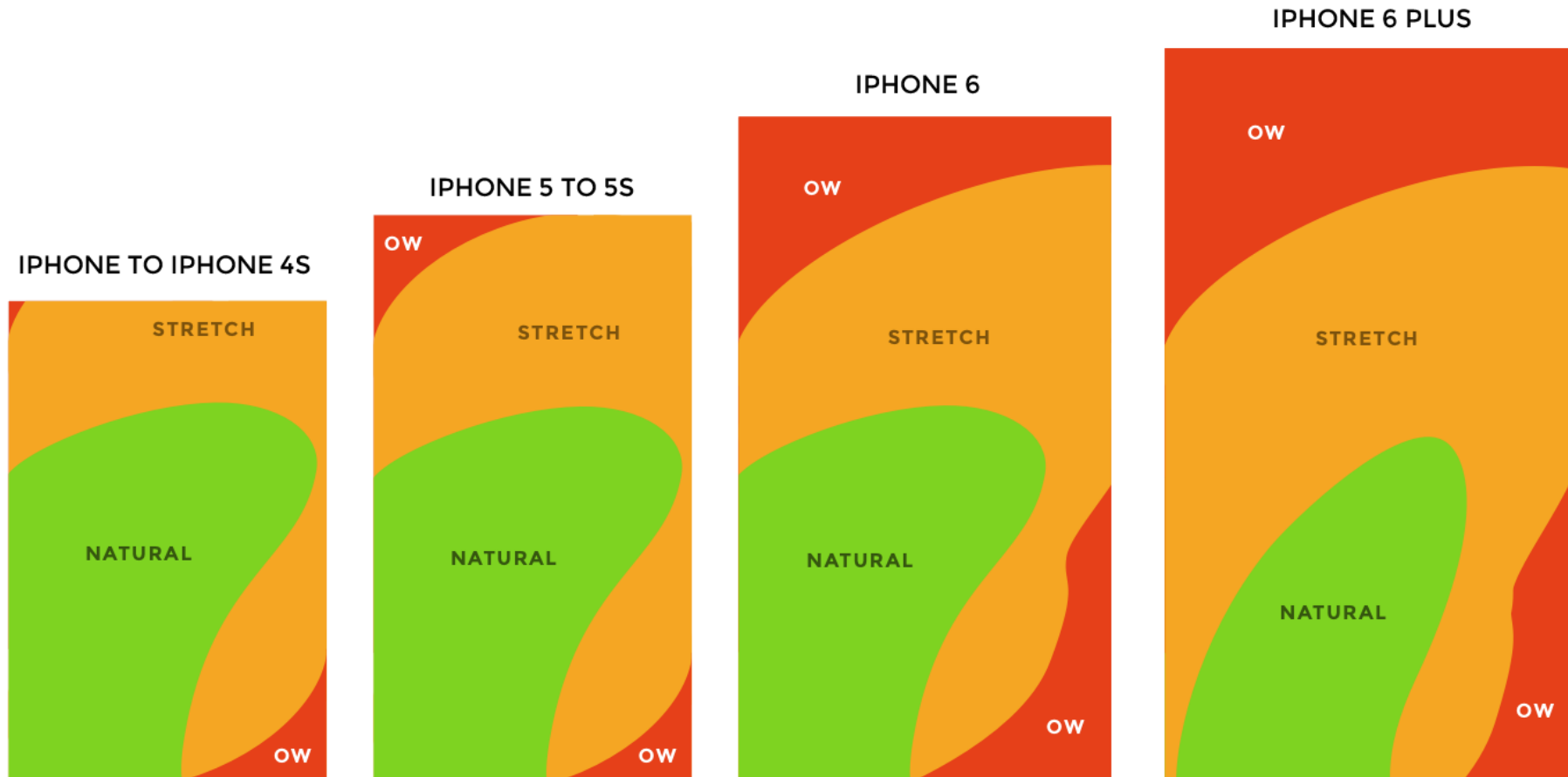
The thumb feels
a little stretched
at these areas.

OW!

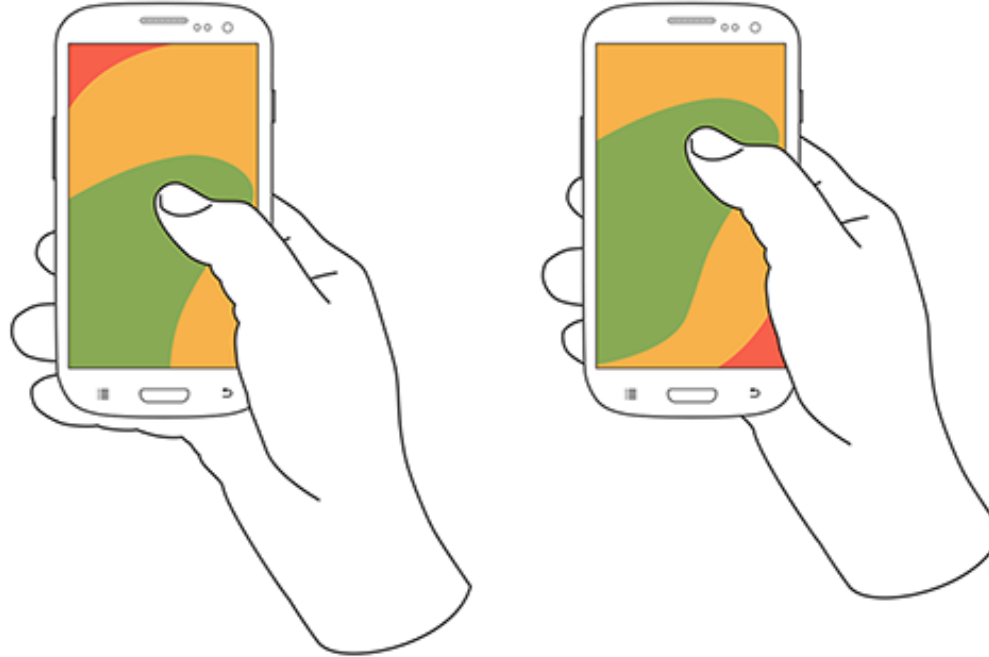
The extended
level of stretch. A
pain to access
this area with
your thumb.

*Right hand holding

Thumb Zone



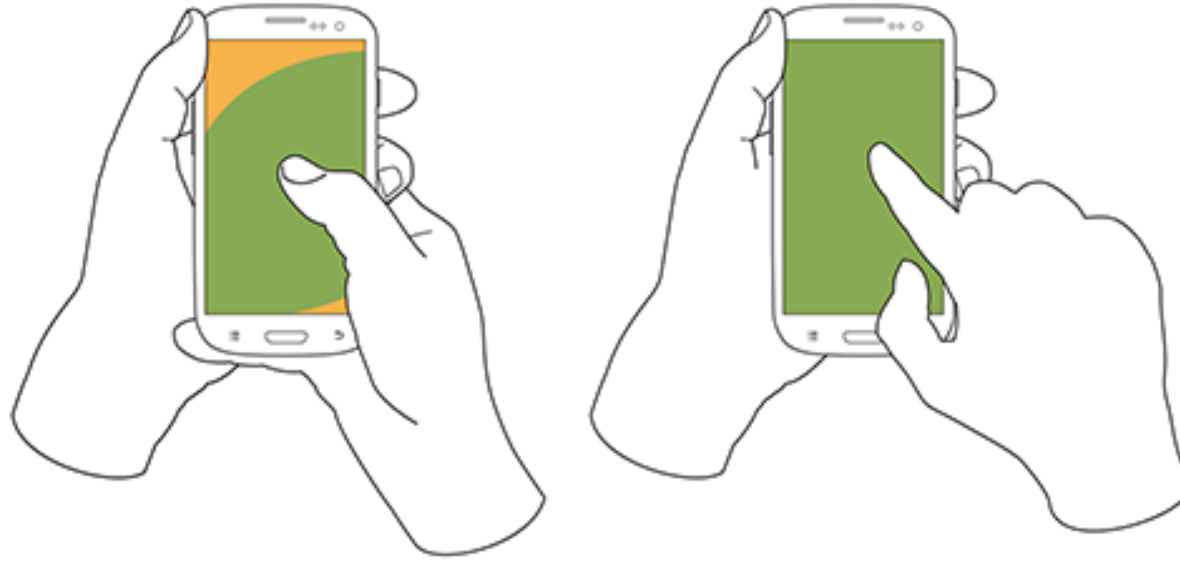
One-Handed Use



The thumb joint is higher in the image on the right. Some users seemed to position their hand by considering the reach they would need.

For example, they would hold the phone so they could easily reach the top of the screen rather than the bottom.

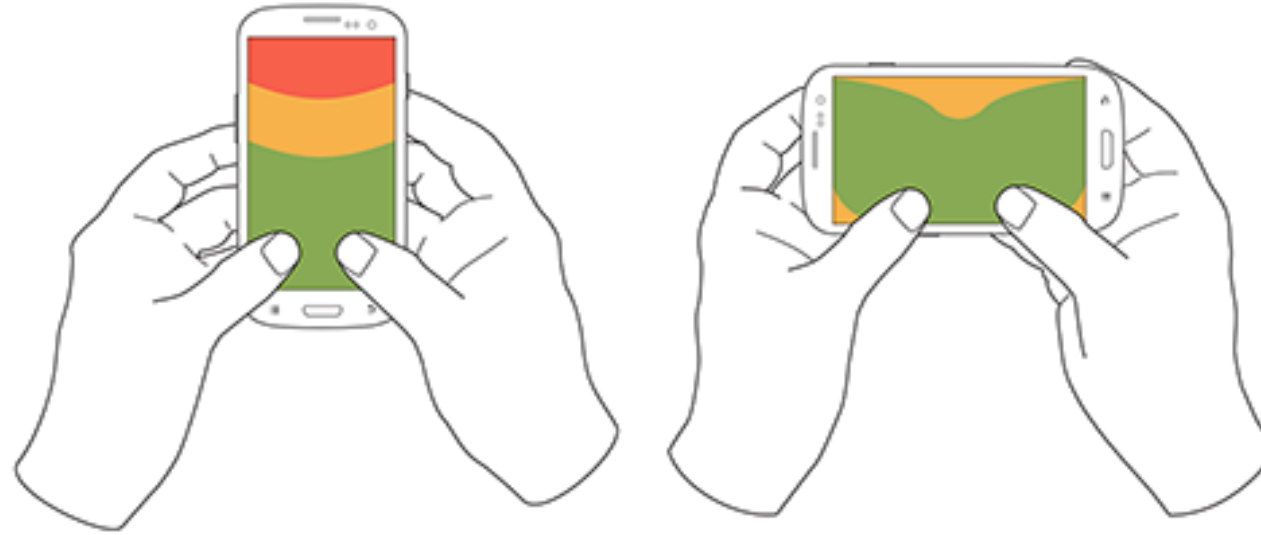
Cradling



With **thumb usage**, users merely added a hand to stabilize the phone for one-handed use.

A smaller percentage of users **use index finger** to interact with the screen. This is similar to the way people use pens with their mobile devices.

Two Handed



Mobile device users often **switched between** two-handed use and cradling, with users typing with both thumbs.

Some users seemed to be adept at tapping the screen with both thumbs or just one thumb. For example, a user might scroll with the right thumb, then tap a link with the left thumb moments later.

Summary of how people hold and interact with mobile phones



49%

one handed



36%

cradled



15%

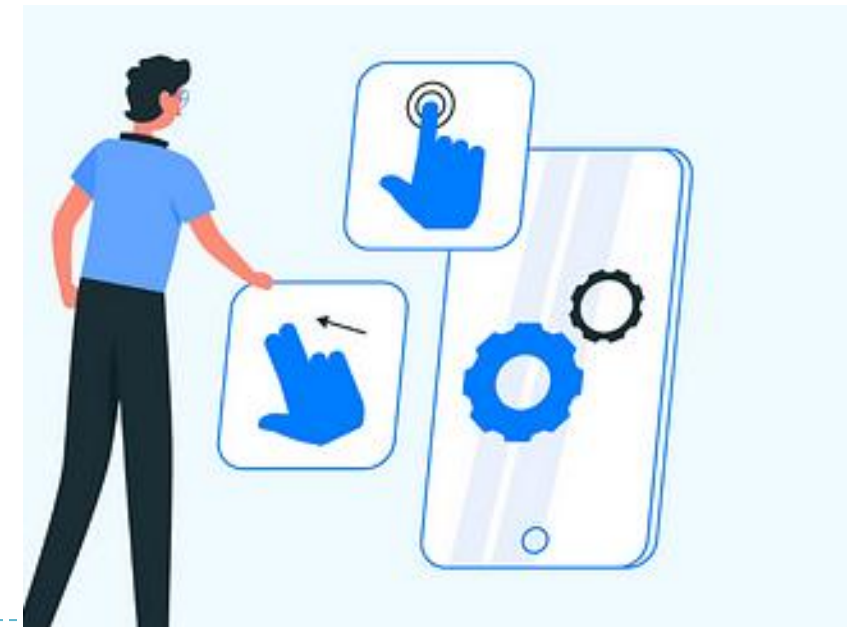
two handed

image source-<https://www.medium.com>

Gestures



- ▶ Each part of a touchscreen *dedicated to **functionality*** precludes the display of content in that area, making **gestures** a crucial component of mobile interaction design.
- ▶ Importance
 - ▶ Easy to Use
 - ▶ Great Onboarding Experience
 - ▶ Intuitive Controls
 - ▶ Saving Space
 - ▶ Content-Focused Mobile Application
 - ▶ More Lively User Experience
 - ▶ Removing the Clutter from the UI

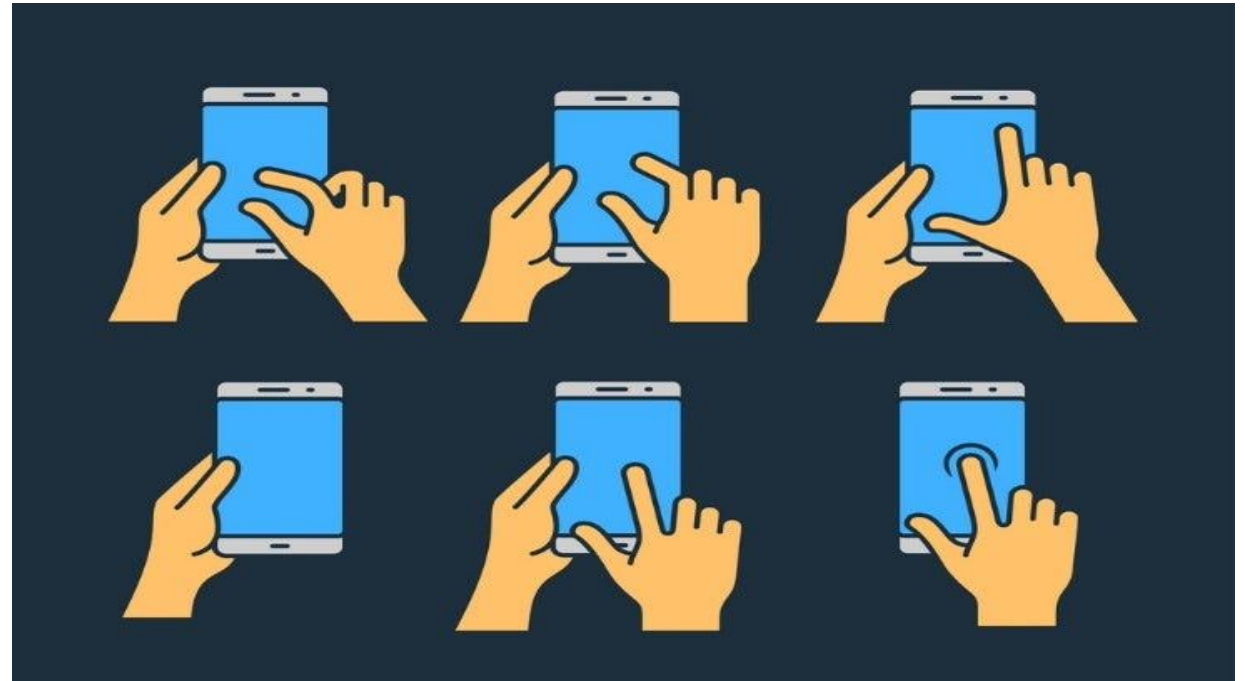


Gestures



▶ There are several popular gestures:

- ▶ Press / Single Tap
- ▶ Long Press / Long Tap
- ▶ Double Press / Double Tap
- ▶ Small Swipe
- ▶ Large Swipe / Fling
- ▶ Pinch / Spread



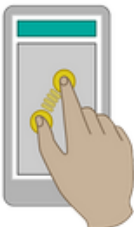
Gestures



Action	iOS	Android	Windows 8
Press 	Selects primary action	Selects primary action	Selects primary action
Long press 	In editable text display magnified view for cursor positioning	Enter data selection mode	Show tooltip without selecting the item.
Double press 	Zoom in or zoom out	Zoom in or zoom out. Also used for text selection	No standard

Gestures



Action	iOS	Android	Windows 8
Small swipe 	Reveal a delete button by sliding horizontally on a table view	No standard	Select objects in a list or grid by sliding perpendicular to panning direction
Large swipe 	Scroll across content	Scroll across content or navigate between views within the same hierarchy	Scroll across content Also used for moving, drawing or writing
Pinch / spread 	Zoom in or zoom out	Zoom in or zoom out	Zoom in or zoom out



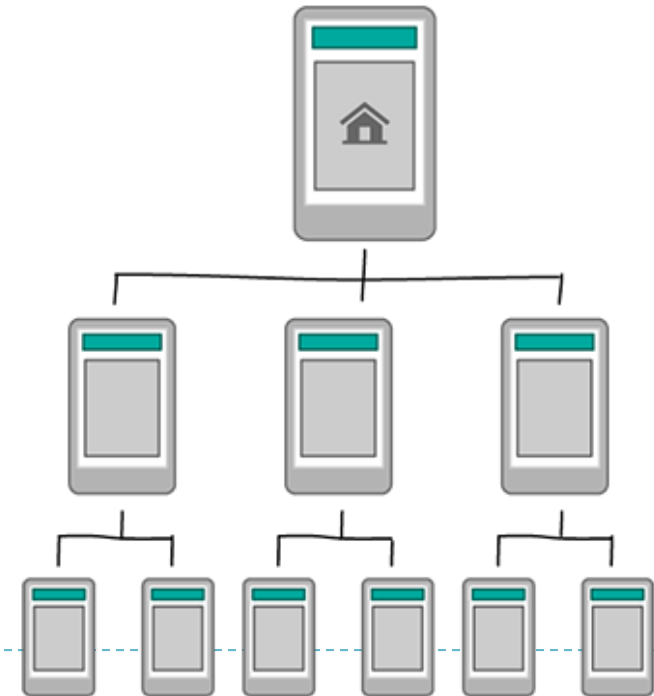
Mobile Information Architecture

- ▶ While the structure of a **responsive site may follow more “standard” patterns**, **native apps, for example, often employ navigational structures that are tab-based.**
- ▶ There’s no “right “way to architect a mobile site or application
- ▶ Some of the most popular patterns:
 - ▶ Hierarchy,
 - ▶ Hub & spoke,
 - ▶ Nested doll
 - ▶ Bento box
 - ▶ Tabbed view,
 - ▶ Filtered view:

Hierarchy



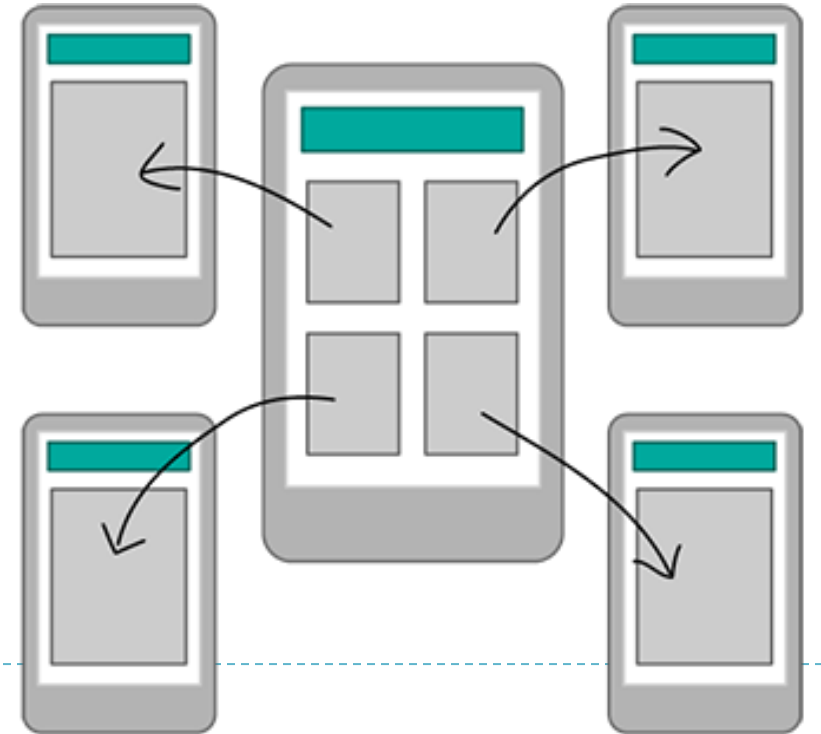
- ▶ a standard site structure with an **index page and a series of sub pages**.
- ▶ If you are designing a **responsive site**, you may be restricted to this.
- ▶ Good for organizing complicated site structures that need to follow a desktop site's structure.



Hub & spoke



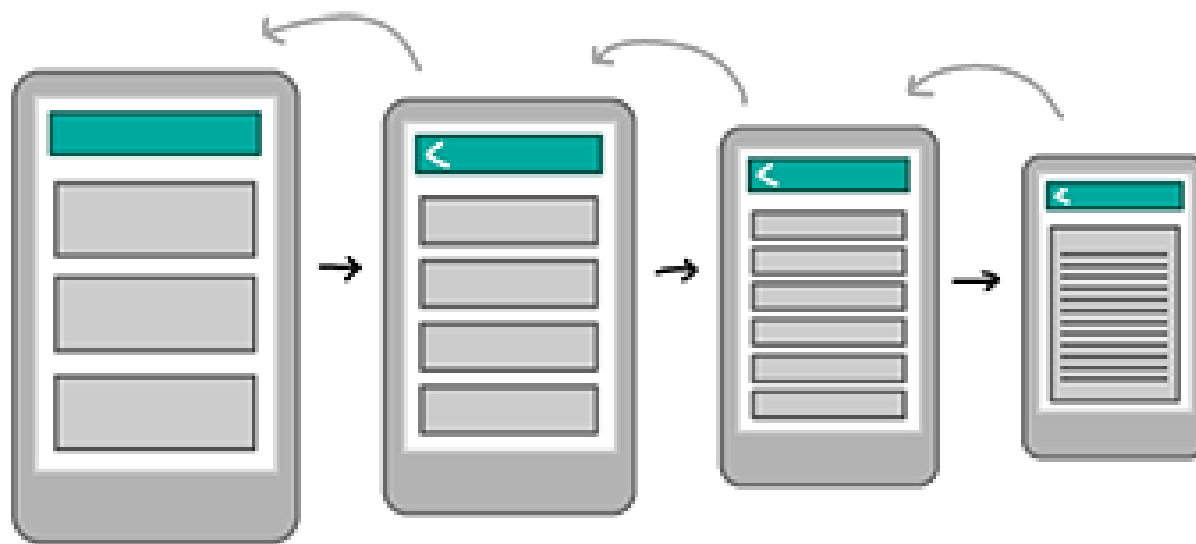
- ▶ A hub and spoke pattern gives you **a central index from which users will navigate out.**
- ▶ Users **can't navigate between spokes** but must return to the hub, instead.
- ▶ Good for multi-functional tools, each with a distinct internal navigation and purpose.





Nested doll

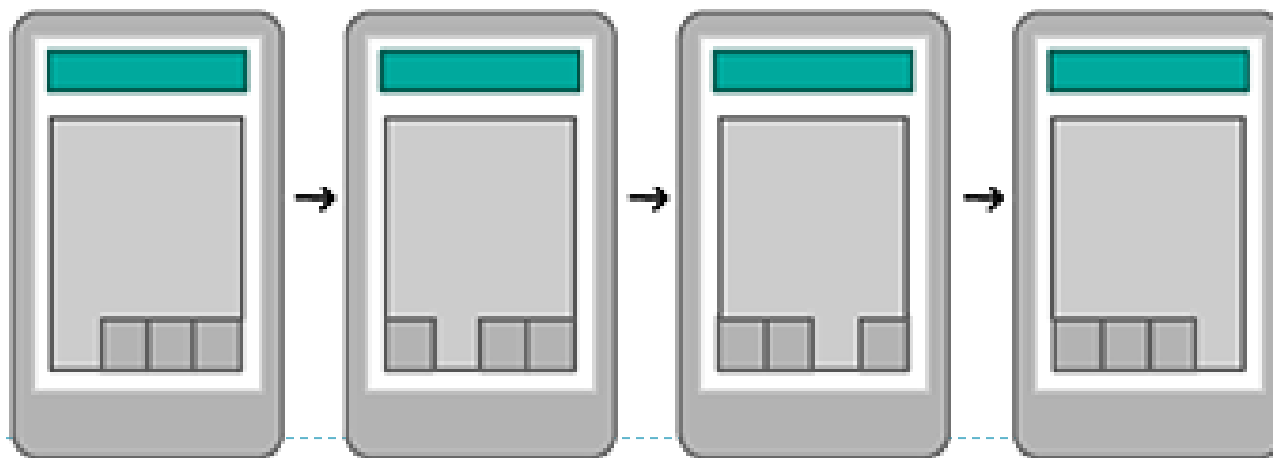
- ▶ The nested doll pattern **leads users in a linear fashion to more detailed content.**
- ▶ Good for apps or sites with **singular or closely related topics.** This can also be used as a sub section pattern inside other parent patterns, such as the standard hierarchy pattern or hub and spoke.





Tabbed view

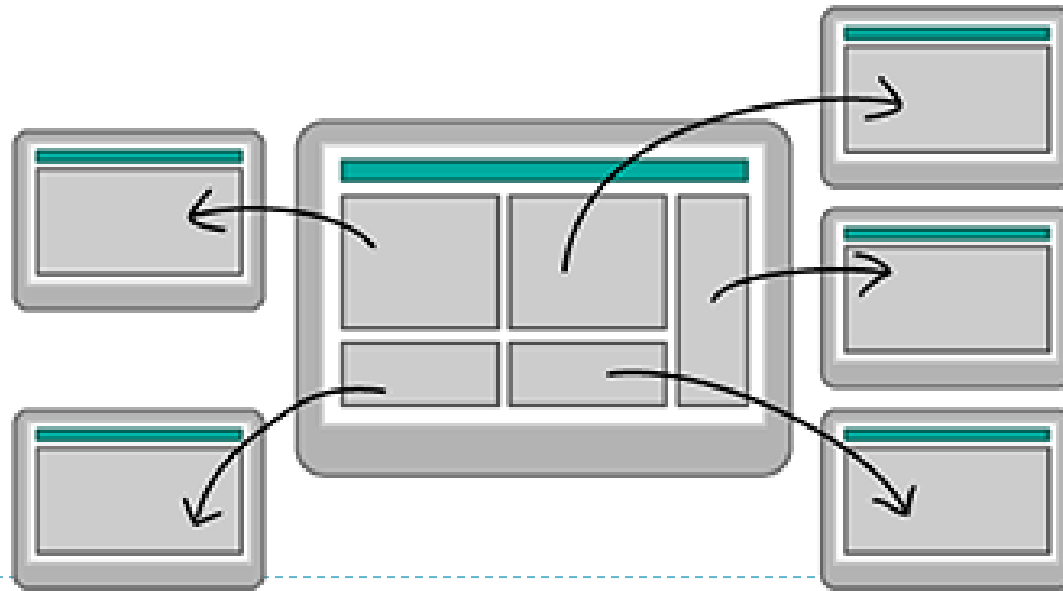
- ▶ A pattern that regular app users will be familiar with.
- ▶ **A collection of sections tied together by a toolbar menu.**
- ▶ Allows the user to quickly scan and understand the complete functionality of the app when it's first opened.
- ▶ Good for tools-based apps with a similar theme. **Multi-tasking.**



Bento Box/Dashboard



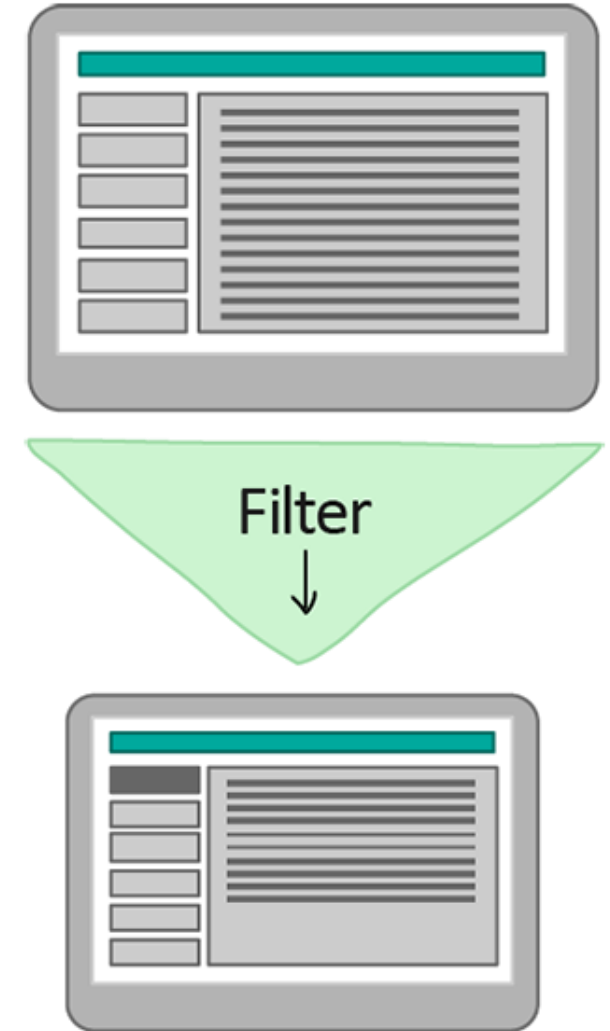
- ▶ Brings **more detailed content directly to the index screen** by using components to display portions of related tools or content.
- ▶ More **suited to tablet** than mobile due to its complexity.
- ▶ Good for multi-functional tools and content-based tablet apps that have a similar theme.



Filtered view



- ▶ A filtered view pattern **allows the user to navigate within a set of data by selecting filter options to create an alternative view.**
- ▶ An excellent way to allow users to explore content in a way that suits them.
- ▶ **Good for apps or sites with large quantities of content**, such as articles, images and videos. (magazine style apps or sites)





DESIGNING FOR CONTEXT





Designing For Context

- ▶ Designing for Mobile Devices should consider **Context**
- ▶ Previously we've already discussed about context-aware mobile application.
- ▶ But what is designing for context anyway?
 - ▶ Context is about understanding human relationship to people, places, and things in the world.



Designing for Context

- ▶ Context enables a user to better understand a concept, person, place, a thing, a situation by adding information to it
- ▶ Context is **important to be designed** into mobile devices as the mobile device users is easily distracted and always expecting a clear and concise information.
- ▶ A mobile web application designed without context will easily **confuse** and **frustrate** mobile users.
 - ▶ (miniaturize vs mobilize!)

Two kinds of context



Context

with a big “C”

VS

context

with a little “c”

Context VS context by example



- ▶ The app works in a specific **context** depending on the current location of the user
- ▶ The app gives **Context** by providing additional info



Context VS context by example

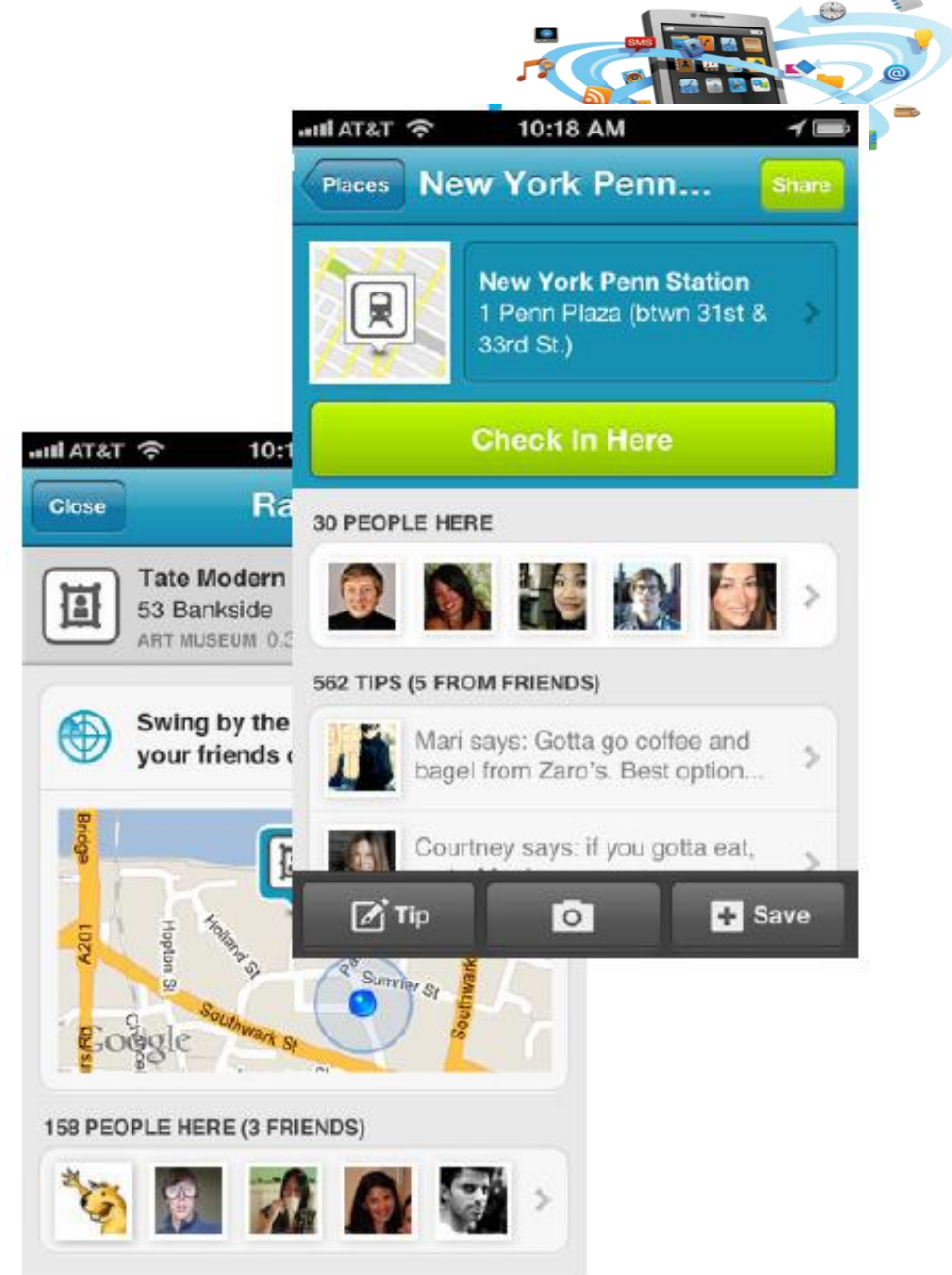
context is a fact

context: where I am

VS

Context: who is here?

Context answers
user's question



<https://foursquare.com>

context with a little “c”

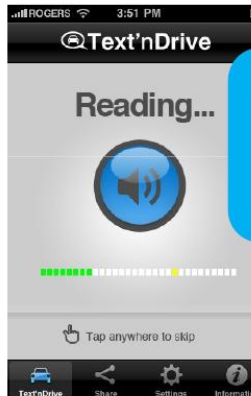


- The mode, medium and environment in which we perform our tasks.

Physical context

In which environment I am

- at home
- on a bus
- in the streets
- in my office
- in a shop



- Connectivity
- Screen size
- Camera
- etc ...

Media context

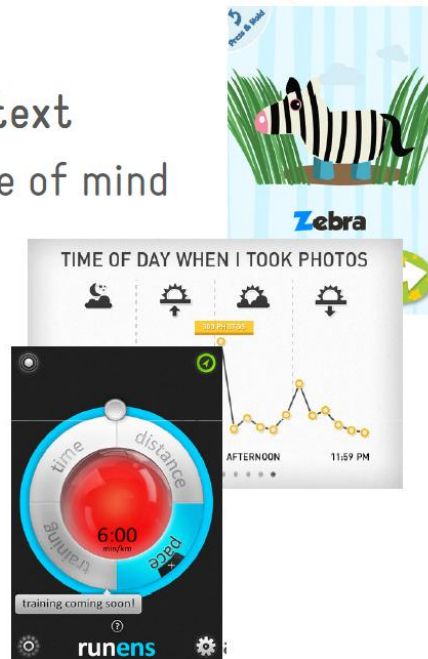
The device I am using



Modal context

My present state of mind

what I am doing right now
How I feel right now





Design for big C

- ▶ User do not care about context (physical, media, modal).
- ▶ They simply care about the app perceived value... **Context!**
- ▶ Good Design means **making the context transparent** to the user
- ▶ We have to create apps that anticipate and solve problems **with little or no action from the user**
- ▶ As a good mobile app developer.... **Design for big “C”**

Designing for Context



Four General Rule-of-Thumb in Designing for Context :

1. **Target users**
2. **Application Use-Case**
3. **Situation and Timing**
4. **Location**



Designing for Context



▶ Application Use-Case

- ▶ Why would the users use the application? To search tickets? Find taxi? Take photos? Share info and ratings? Get status updates?
- ▶ Why would you as a student use the application?



Designing for Context



► Situation and Timing

- ▶ When will they use the application?
- ▶ Will they be alone?
- ▶ Will they be with friends?
- ▶ During tense situation?
- ▶ During happy moments?



Designing for Context



► Location

- Where would they typically use the application?
- Outdoor?
- Indoor?
- Within the mall?
- At the bus station?
- At a diner or café?



Idea for developing Context-Aware Application (Step-by-Step Guide)



- ▶ **Step 1:** Think about existing application are out there that can benefit from context.
- ▶ **Step 2:** Think about the signals that will deliver context to the apps that you design

LOCATION

ACTIVITY

walk, run, in car,
on bike, still

CURRENT WEATHER

UV Index,
temperature,
humidity, rain,
sunrise & sunset

WEATHER FORECAST

10 day forecast

Contextual Signals on iOS & Android



Idea for developing Context-Aware Application (Step-by-Step Guide)



- ▶ **Step 3:** Think about how you could leverage different time periods of data to interact with the user

**ON DEMAND /
RIGHT NOW**

**HISTORY /
USAGE OVER
TIME**

**PLANNING &
RESEARCH**

3 types of contextual interactions that users can have with contextual experiences

Idea for developing Context-Aware Application (Step-by-Step Guide)



► **Step 4:** Combine Steps 1 to 3 to produce new context

FITNESS

runtastic warns it will rain in a few days when you have a run scheduled... you can also glean patterns of activity from history

MUSIC

think about how UI changes for music app based on activity. e.g.: in car, don't show playlist editor, only show simple playback controls

ASSIST

based on loc and/or activity, enable/disable features on phone. e.g.: disable ringer @ home. e.g.: send calls to voicemail while in vehicle

WEATHER

collect prefs from user for temp ranges, rate of change, rain, UV index, for alerts... give recommendations, like bring an umbrella, or jacket

CLOTHING

based on monthly collection of outfits, use weather data to suggest what outfit the user wears that day

YELP

give suggestions of places based on weather, e.g.: patio seating is relevant when it's not raining or super cold or hot

UBER

going to rain later today, so beware of surge pricing and maybe leave sooner or later

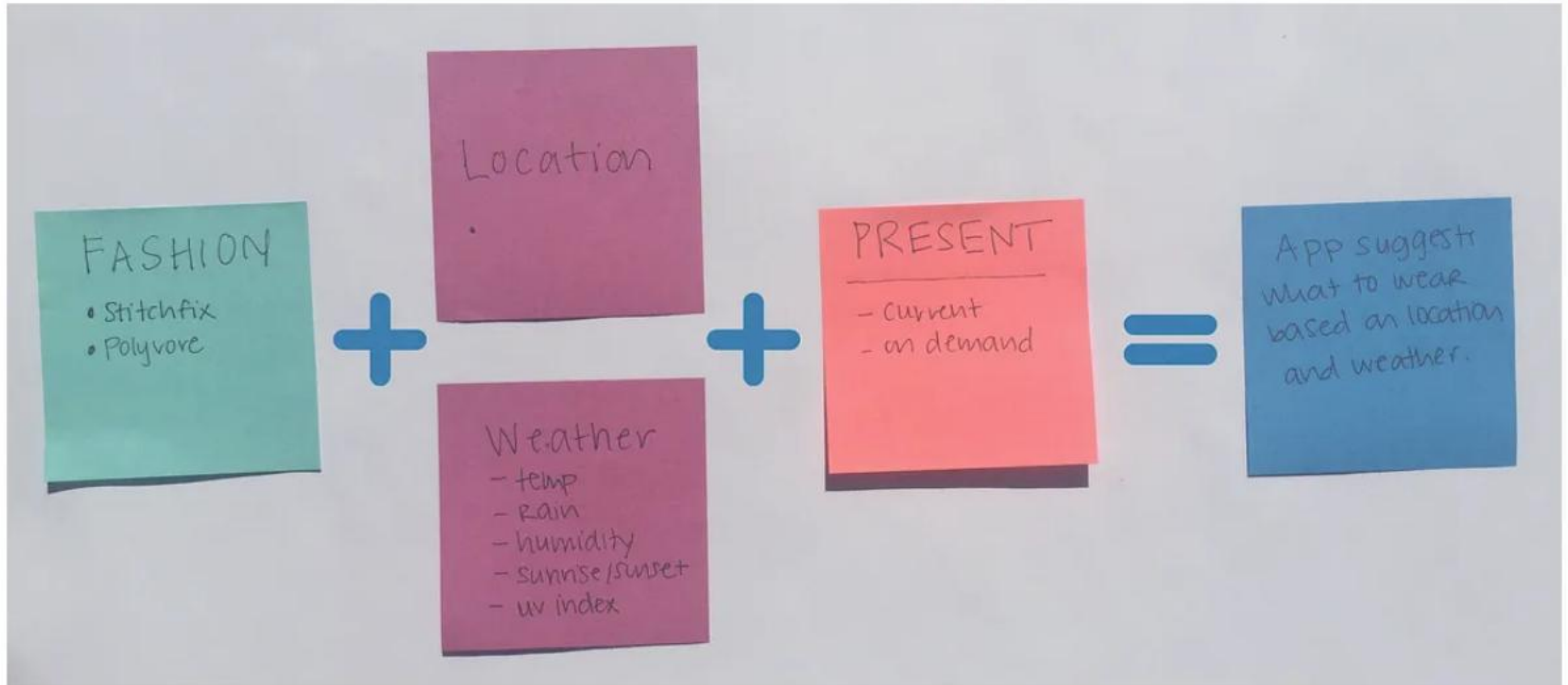
SMART CLOCK

wake me up early because it snowed last night

Step-by-Step Context-Aware Workflow



brainstorming the possibilities



Putting it all together for this app concept

END